

A strict failure of reversed double semi-polarity

Candidate counterexample for arXiv:1308.0974

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Status: candidate counterexample/full negative answer, likely valid. In the smooth, strictly convex space ℓ_4^2 , an explicit convex body M with the origin in its interior satisfies

$$M \subsetneq (M^\circ)_\circ.$$

More generally, the reversed double semi-polar is the inverse image under the norm-duality map of a convex hull.

1 Source and open question

The source is A. G. Horváth, Z. Lángi, and M. Spirova, *Semi-inner products and the concept of semi-polarity*, arXiv:1308.0974, subsequently published in *Results in Mathematics* 71 (2017), 127–144 [1].

Let $(X, \|\cdot\|)$ be finite dimensional, smooth, and strictly convex, and let $[\cdot, \cdot]$ be its induced semi-inner product. For $M \subset X$ the source defines

$$M_\circ = \{x : [m, x] \leq 1 \text{ for all } m \in M\}, \quad M^\circ = \{x : [x, m] \leq 1 \text{ for all } m \in M\}.$$

It proves $(M_\circ)^\circ = M$ whenever M is a convex body containing the origin in its interior. Section 6, Question 2 asks whether the reversed identity $M = (M^\circ)_\circ$ is also true.

2 The structural obstruction

Write

$$F : X \longrightarrow X^*, \quad F(y)(x) = [x, y].$$

Under the standing smoothness and strict-convexity assumptions, the generalized Riesz representation theorem recalled by the source says that F is a bijection. It is positively and negatively homogeneous but need not be additive.

Theorem 1 (Convex-hull formula). *For every convex body $M \subset X$ containing the origin in its interior,*

$$(M^\circ)_\circ = F^{-1}(\text{conv } F(M)). \tag{1}$$

Consequently,

$$M = (M^\circ)_\circ \iff F(M) \text{ is convex.}$$

Proof. For $A \subset X^*$ and $C \subset X$, use the ordinary one-sided polars

$$A^* = \{x \in X : \varphi(x) \leq 1 \text{ } (\varphi \in A)\}, \quad C^* = \{\varphi \in X^* : \varphi(x) \leq 1 \text{ } (x \in C)\}.$$

each symplectic form for any norm. Note that, for normed spaces, the counterpart of an orthogonal basis is a so-called *Auerbach basis*, which is a basis containing pairwise normal unit vectors with respect to the norm. This leads to the following question.

Problem 1. *Prove or disprove that if $\|\cdot\|_a = \|\cdot\|$ with respect to any symplectic form with a polar decomposition into an Auerbach basis of $(\mathbb{X}, \|\cdot\|)$, then $(\mathbb{X}, \|\cdot\|)$ is Euclidean.*

Problem 2. *Characterize the norms $\|\cdot\|$ satisfying $\|\cdot\|_a = \|\cdot\|$ with respect to some symplectic form.*

Note that the normality map J depends on the choice of the symplectic form $\langle \cdot, \cdot \rangle$ on $\mathbb{X}$.

Question 1. *Let $(\mathbb{X}, \|\cdot\|)$ be of dimension $2n > 2$. Do there exist symplectic forms with respect to which JS is the isoperimetrix of $(\mathbb{X}, \|\cdot\|)$ in the sense of Busemann or Holmes-Thompson?*

For these two concepts of isoperimetrices see, e.g. Chapter 5 of [23], or [18].

Question 2. *We have shown in Theorem 5.2 that for any $M \in \mathfrak{X}_o$, we have $(M_o)^\circ = M$. Clearly, $M \subseteq (M^\circ)_o$ also holds. Is it true that $M = (M^\circ)_o$?*

Figure 1: The open question on page 14 of the source PDF.

The definitions give

$$M^\circ = F(M)^*, \quad (M^\circ)_o = \{y : F(y) \in (M^\circ)^*\}.$$

The ordinary bipolar separation argument states that $(A^*)^* = \text{cl conv}(A \cup \{0\})$. Indeed, the inclusion from right to left is immediate, while a point outside that closed convex hull is strictly separated from it by evaluation at some $x \in X$; rescaling x puts it in A^* and excludes the point from $(A^*)^*$.

Here $0 \in M$ and $F(0) = 0$. Moreover F is continuous: if $y_k \rightarrow y$, then $\|F(y_k)\|_* = \|y_k\|$; every convergent subsequence of $F(y_k)$ has a limit norming y , and uniqueness of the norming functional forces that limit to be $F(y)$. Thus $F(M)$ is compact, and in finite dimensions its convex hull is compact. Applying the bipolar identity to $F(M)$ yields (1). Since F is bijective, equality with M holds exactly when $\text{conv } F(M) = F(M)$. $\square$

The formula pinpoints why the source's proved order works while the reversed one may fail: the nonlinear duality map need not carry a convex body to a convex set.

3 An explicit counterexample in ℓ_4^2

Let $X = \mathbb{R}^2$ with the ℓ_4 norm. This norm is smooth and strictly convex. Its standard induced semi-inner product is

$$[x, z]_4 = \frac{x_1 z_1^3 + x_2 z_2^3}{\|z\|_4^2} \quad (z \neq 0), \quad [x, 0]_4 = 0. \quad (2)$$

Let B_2 denote the Euclidean unit disk, let e_1, e_2 be the coordinate vectors, and set

$$M = \text{conv} \left(\frac{1}{10} B_2 \cup \{e_1, e_2\} \right), \quad y = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right). \quad (3)$$

Theorem 2. *The set M in (3) is a convex body with 0 in its interior, and*

$$y \in (M^\circ)_\circ \setminus M.$$

In particular, Question 2 has a negative answer.

Proof. The inclusion $(1/10)B_2 \subset M$ makes M a compact convex body with 0 in its interior. Consider the linear functional $L(z) = z_1 + z_2$. Its maximum on $(1/10)B_2$ is $\sqrt{2}/10 < 1$, while $L(e_1) = L(e_2) = 1$. Therefore $L \leq 1$ throughout M . But

$$L(y) = \sqrt{2} > 1,$$

so $y \notin M$.

Now let $x = (x_1, x_2) \in M^\circ$. Since $e_1, e_2 \in M$, the definition of the right semi-polar and (2) give

$$x_1 = [x, e_1]_4 \leq 1, \quad x_2 = [x, e_2]_4 \leq 1.$$

For the point y in (3), one has $\|y\|_4^2 = 2^{-1/2}$ and $y_1^3 = y_2^3 = 2^{-3/2}$. Hence

$$[x, y]_4 = \frac{x_1 + x_2}{2} \leq 1.$$

This inequality holds for every $x \in M^\circ$, which is precisely the condition $y \in (M^\circ)_\circ$. Together with $y \notin M$, it proves the strict inclusion. $\square$

The same point can be read through Theorem 1: formula (2) gives $F(e_1) = (1, 0)$, $F(e_2) = (0, 1)$, and $F(y) = (1/2, 1/2)$. Thus $F(y)$ lies on the segment joining two points of $F(M)$ even though y lies outside M .

4 Verification, scope, and novelty

The proof is exact and has no numerical dependency. It uses the source's one-sided inequalities exactly as written; in particular no symmetry of M or of the semi-inner product is assumed. The auxiliary Euclidean disk only ensures that 0 is an interior point of M ; the semi-inner product is induced throughout by ℓ_4^2 . The complete convention and hypothesis audit is recorded in the companion `verification.md`.

This is a full negative answer to Question 2 and also supplies a criterion for equality for each individual body. It does not address the source's other questions about antinorms, isoperimetricity, or odd-dimensional analogues.

Before promotion, the run indexes were searched using arXiv:1308.0974, the exact title, semi-polarity, right and left semi-polars, and the displayed double operation. Bounded web searches on 11 August 2026 used the exact question, close variants, the article title and DOI, and citation-oriented queries. They found the source and later papers citing it for background, but no answer to Question 2 and no matching ℓ_4^2 construction. Novelty confidence is moderate because citation/search coverage is bounded.

References

- [1] Á. G. Horváth, Z. Lángi, and M. Spirova, *Semi-inner products and the concept of semi-polarity*, Results Math. 71 (2017), 127–144; arXiv:1308.0974. <https://arxiv.org/abs/1308.0974>.